THE HOMOTOPY CATEGORY OF THE DIFFERENTIAL GRADED CATEGORY OF SITUATION ABC IS PRE-TRIANGULATED

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-dga/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the TR1 verification from the admissible sequence and cone triangles, the full TR2 argument with the three-row sign-change diagram and both directions, and the TR3 reduction to cone triangles with the completing morphism from lemma-cone are all retained verbatim.

Editorial note. The label lemma-analogue-homotopy-category-pre-triangulated names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Distinguished triangles from admissible short exact sequences and cones. Same chapter, Lemmas lemma-cone-homotopy and lemma-cone-rotate-isom. Identifies distinguished triangles with cone triangles up to isomorphism and gives the rotated cone triangle; these two facts are what all three axioms are reduced to.
- Restated axiom (C) under rotation. Same chapter, Lemma lemma-restate-axiom-c. Closes the converse half of TR2: the rotated cone triangle with sign-changed maps is distinguished.

Source reference. Original reference 08C5: sources/stacks-dga/pinned-original-source.tex, source lines 3779–3779. Prerequisite pointer invoked at proof line(s) 4569; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09QU: sources/stacks-dga/pinned-original-source.tex, source lines 4326–4326. Prerequisite pointer invoked at proof line(s) 4515; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09QS: sources/stacks-dga/pinned-original-source.tex, source lines 4167–4167. Prerequisite pointer invoked at proof line(s) 4526; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original reference 09QR: sources/stacks-dga/pinned-original-source.tex, source lines 4148–4148. Prerequisite pointer invoked at proof line(s) 4556; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09QJ: sources/stacks-dga/pinned-original-source.tex, source lines 3762–3762. Prerequisite pointer invoked at proof line(s) 4497; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 09QJ the homotopy category $K(\mathcal{A})$ with its natural translation functors and distinguished triangles is a pre-triangulated category.*

Proof. We will verify each of TR1, TR2, and TR3.

Proof of TR1. By definition every triangle isomorphic to a distinguished one is distinguished. Since

$$x \xrightarrow{1_x} x \longrightarrow 0$$

is an admissible short exact sequence, $(x, x, 0, 1_x, 0, 0)$ is a distinguished triangle. Moreover, given a morphism $\alpha : x \to y$ in $\mathrm{Comp}(\mathcal{A})$, the triangle given by $(x, y, c(\alpha), \alpha, i, -p)$ is distinguished by Lemma 09QU.

Proof of TR2. Let $(x, y, z, \alpha, \beta, \gamma)$ be a triangle and suppose $(y, z, x[1], \beta, \gamma, -\alpha[1])$ is distinguished. Then there exists an admissible short exact sequence $0 \to x' \to y' \to z' \to 0$ such that the associated triangle $(x', y', z', \alpha', \beta', \gamma')$ is isomorphic to $(y, z, x[1], \beta, \gamma, -\alpha[1])$. After rotating, we conclude that $(x, y, z, \alpha, \beta, \gamma)$ is isomorphic to $(z'[-1], x', y', \gamma'[-1], \alpha', \beta')$. By Lemma 09QS, we deduce that $(z'[-1], x', y', \gamma'[-1], \alpha', \beta')$ is isomorphic to $(z'[-1], x', c(\gamma'[-1]), \gamma'[-1], i, p)$. Composing the two isomorphisms with sign changes as indicated in the following diagram:

$$\begin{array}{ccccccc}
x & \xrightarrow{\alpha} & y & \xrightarrow{\beta} & z & \xrightarrow{\gamma} & x[1] \\
\downarrow & & \downarrow & & \downarrow & & \downarrow \\
z'[-1] & \xrightarrow{-\gamma'[-1]} & x & \xrightarrow{\alpha'} & y' & \xrightarrow{\beta'} & z' \\
{\scriptstyle -1_{z'[-1]}}\downarrow & & \| & & \downarrow & & \downarrow{\scriptstyle -1_{z'}} \\
z'[-1] & \xrightarrow{\gamma'[-1]} & x & \xrightarrow{\alpha'} & c(\gamma'[-1]) & \xrightarrow{-p} & z'
\end{array}$$

We conclude that $(x, y, z, \alpha, \beta, \gamma)$ is distinguished by Lemma 09QU (2). Conversely, suppose that $(x, y, z, \alpha, \beta, \gamma)$ is distinguished, so that by Lemma 09QU (1), it is isomorphic to a triangle of the form $(x', y', c(\alpha'), \alpha', i, -p)$ for some morphism $\alpha' : x' \to y'$ in $\mathrm{Comp}(\mathcal{A})$. The rotated triangle $(y, z, x[1], \beta, \gamma, -\alpha[1])$ is isomorphic to the triangle $(y', c(\alpha'), x'[1], i, -p, -\alpha[1])$ which is isomorphic to $(y', c(\alpha'), x'[1], i, p, \alpha[1])$. By Lemma 09QR, this triangle is distinguished, from which it follows that $(y, z, x[1], \beta, \gamma, -\alpha[1])$ is distinguished.

Proof of TR3: Suppose $(x, y, z, \alpha, \beta, \gamma)$ and $(x', y', z', \alpha', \beta', \gamma')$ are distinguished triangles of $\mathrm{Comp}(\mathcal{A})$ and let $f : x \to x'$ and $g : y \to y'$ be morphisms such that $\alpha' \circ f = g \circ \alpha$. By Lemma 09QU, we may assume that $(x, y, z, \alpha, \beta, \gamma) = (x, y, c(\alpha), \alpha, i, -p)$ and $(x', y', z', \alpha', \beta', \gamma') = (x', y', c(\alpha'), \alpha', i', -p')$. Now apply Lemma 08C5 and we are done. □